

DEVELOPMENT OF MICRO-LEARNING ACTIVITIES BASED ON CONNECTIVIST PRINCIPLES TO ENHANCE CRITICAL THINKING SKILLS OF UNDERGRADUATE PHYSICS STUDENTS

**A SYNOPSIS SUBMITTED TO
SHIVAJI UNIVERSITY, KOLHAPUR**

**FOR THE DEGREE OF
B.Ed.-M.Ed. (Integrated) Three Years Programme**

**SUBJECT: EDUCATION
FACULTY: INTERDISCIPLINARY STUDIES**

**SUBMITTED BY
Mr. Kishore Shyam
PRN: 2023085001**

**UNDER THE GUIDANCE OF
Dr. (Ms.) Sankpal Rupali Uttam
Associate Professor
Department of Education
Shivaji University, Kolhapur**

**DEPARTMENT OF EDUCATION
SHIVAJI UNIVERSITY, KOLHAPUR**

April – 2026

CONTENTS

Sr. No.	Title	Page No.
1.	Introduction	1
2.	Need of the Study	1
3.	Statement of the Problem	2
4.	Variables of the Study	2
5.	Definition of the Terms	3
6.	Review of Related Literature	4
7.	Objectives of the Study	6
8.	Assumptions of the Study	7
9.	Hypotheses of the Study	7
10.	Significance of the Study	8
11.	Scope of the Study	8
12.	Delimitations of the Study	9
13.	Limitations of the Study	10
14.	Procedure of the Research	11
15.	Sampling Design and Sample	12
16.	Tools for Data Collection	17
17.	Development of the Micro-learning Programme	18
18.	Analysis, Interpretation of Data and Results	20
19.	Recommendations and Implications	23
20.	Suggestions for Further Research	24

1 Introduction

Undergraduate physics education presents a persistent challenge: students frequently learn to arrive at correct answers without developing genuine conceptual understanding. Abstract content, mathematical demands, and examination pressure together tend to reinforce surface-level memorization rather than deeper engagement with ideas.

The National Education Policy (NEP) 2020 identifies critical thinking as a central outcome of higher education in India (Ministry of Education, 2020). Yet undergraduate classrooms have been slow to reflect this priority. Lecture-based instruction continues to dominate, and although students have widespread access to smartphones and digital platforms, these resources are rarely integrated into the learning process in a structured manner.

Micro-learning — consisting of short, focused instructional units — has been identified as a promising approach for supporting learning in complex subject areas (Giurgiu, 2017; Leong et al., 2021). When designed around connectivist principles, which position learning as occurring through networks, peer interaction, and distributed knowledge sources (Downes, 2007; Siemens, 2005), such approaches can facilitate more active cognitive engagement.

Notwithstanding these developments, critical thinking remains underdeveloped among undergraduate physics students (Tiruneh et al., 2017). Instructional approaches that move beyond content delivery and actively support analysis, evaluation, and reasoning are therefore required.

The present study was initiated through a needs analysis survey of undergraduate physics teachers, aimed at identifying topics that are most difficult to teach for conceptual understanding based on classroom experience. A connectivist micro-learning programme was subsequently designed and implemented around the identified topic. The study examines whether participation in this programme leads to measurable improvement in the critical thinking skills of undergraduate physics students.

2 Need of the Study

The National Education Policy (NEP) 2020 calls for higher-order cognitive skills — critical thinking, problem-solving, and independent reasoning — as central outcomes of undergraduate education in India (Ministry of Education, 2020). The gap between this expectation and actual classroom practice remains considerable.

This gap is particularly evident in physics education. The discipline demands abstract reasoning, mathematical fluency, and the ability to apply concepts across unfamiliar contexts. However, undergraduate instruction often emphasizes procedural problem-solving and examination preparation. As a result, students may be able to arrive at correct answers without fully understanding the underlying concepts or reasoning pro-

cesses.

Lecture-based teaching further reinforces this pattern. When learning is primarily passive — involving listening, note-taking, and reproduction — students have limited opportunities to analyse, question, or connect ideas. Consequently, the development of critical thinking skills remains constrained.

Micro-learning — which involves breaking complex content into shorter, focused units — has been identified as a promising approach for improving engagement and retention (Giurgiu, 2017; Leong et al., 2021). Connectivist principles complement this by emphasizing learning through networks, peer interaction, and distributed knowledge sources (Downes, 2007; Siemens, 2005). Together, these approaches offer a pedagogically sound and practically feasible alternative to traditional instruction.

However, such approaches are rarely implemented in a structured manner within undergraduate physics education. The present study addresses this gap by designing, implementing, and evaluating a connectivist micro-learning programme aimed at enhancing the critical thinking skills of undergraduate physics students.

3 Statement of the Problem

The problem addressed in the present study is formally stated as follows:

“Development of Micro-learning Activities Based on Connectivist Principles to Enhance Critical Thinking Skills of Undergraduate Physics Students.”

4 Variables of the Study

The variables of the present study were identified from the research title, *“Development of Micro-learning Activities Based on Connectivist Principles to Enhance Critical Thinking Skills of Undergraduate Physics Students,”* and are defined as follows:

1. **Independent Variable:** The term “micro-learning programme based on connectivist principles” in the research title was operationalised as the connectivist micro-learning programme developed by the researcher and implemented for the experimental group, while the control group continued with regular classroom instruction.
2. **Dependent Variable:** The term “enhance critical thinking skills” in the research title was operationalised as the critical thinking performance of undergraduate physics students, measured through the General Critical Thinking Assessment (Paper 1) and the Physics Domain-Specific Critical Thinking Assessment (Paper 2).

3. **Control Variables:** Variables such as academic level (First-Year B.Sc.), subject (Physics), and syllabus (NEP 2020–aligned curriculum) were kept constant across groups.
4. **Extraneous Variables:** Factors such as prior knowledge, motivation, attendance, and access to digital resources, which were not fully controlled, may have influenced the outcomes of the study.

5 Definition of Terms

The key terms used in the present study are defined operationally as follows:

- **Development:** In the present study, development refers to the systematic process followed to design the connectivist micro-learning programme. This process included identifying relevant connectivist principles through literature review and expert validation; conducting a needs analysis survey among undergraduate physics teachers to identify topics that are challenging to teach, difficult for students to learn, and require instructional support; identifying Electrostatics as the focus area based on the findings of the teacher survey; developing critical thinking assessment tools; conducting a pilot study to refine these tools; and designing and validating the micro-learning programme based on expert feedback.
- **Micro-Learning Activities Based on Connectivist Principles:** In the present study, this refers to a series of short-duration, structured learning sessions delivered through the Slack digital platform for teaching the topic of Electrostatics. These activities involved guided discussion, peer interaction, and sharing of learning resources, facilitated by the researcher, and were implemented over the duration of the intervention as part of the micro-learning programme.
- **Enhancement of Critical Thinking Skills:** In the present study, this refers to an improvement in students' performance on the critical thinking assessment, as indicated by the difference between pre-test and post-test scores. This improvement is determined using the researcher-developed assessment tools, namely the General Critical Thinking Assessment (Paper 1) and the Physics Domain-Specific Critical Thinking Assessment (Paper 2), administered before and after the implementation of the micro-learning programme.
- **Undergraduate Physics Students:** In the present study, this refers to First-Year Bachelor of Science (F.Y. B.Sc.) students who have physics as a compulsory subject in their programme, enrolled in affiliated colleges within Kolhapur City and in relevant departments of Shivaji University, Kolhapur, where undergraduate physics is taught, under the NEP 2020–aligned curriculum.

6 Review of Related Literature

A review of related literature was undertaken to examine existing research on micro-learning, connectivism, and critical thinking in science and physics education. Given the scope of the present synopsis, a focused selection of relevant studies has been reviewed.

Research on micro-learning consistently points to its value in improving engagement and knowledge retention. Shail and Shail (2019) found that mobile-based micro-learning reduces cognitive overload and supports spaced repetition, both of which aid retention. Diniyatushoaliha (2024) reported that structured digital modules can develop analytical skills including interpretation and evaluation. Sastry (2021) observed that combining mobile learning with metacognitive strategies produced significant gains in higher-order thinking. Across these studies, however, structured applications of micro-learning within higher education—particularly in demanding subjects such as physics—remain limited.

Connectivism frames learning as a networked process dependent on interaction, collaboration, and access to distributed knowledge sources. Siemens (2005) positioned connectivism as a framework suited to digital-age learning environments. Joshi (2021) found that connectivist strategies improved critical thinking and problem-solving in science education, and Al-Mutairi and Mubayrik Bin (2021) observed that connectivist settings support ongoing learning through peer exchange and shared resources. Despite this theoretical grounding, studies that translate connectivist principles into structured instructional practice remain limited.

In physics education specifically, critical thinking does not develop without deliberate instructional support. Deshpande (2020) found that targeted interventions can substantially improve critical thinking even among students with no prior exposure to such training. Binoj (2022) reported that ICT-based and blended learning approaches outperform traditional instruction in developing reasoning abilities. Ramadhani et al. (2024) noted that active learning strategies, while effective, face particular difficulty when applied to abstract physics content. Maries and Singh (2023) further highlighted the role of metacognitive scaffolding in supporting physics problem-solving.

To provide an overview of the scope and distribution of the reviewed literature, the selected studies were classified based on their thematic focus and geographical context. The distribution of the reviewed studies is presented in Table 2.

Table 2
Distribution of Reviewed Research Studies

Sr. No.	Focus Area	Indian Studies	Studies Abroad
1	Micro-learning Approaches	3	5
2	Connectivism & Networked Learning	4	9
3	Critical Thinking & Problem-Based Learning	8	17
4	ICT, Blended & E-Learning in Science	21	18
5	Constructivist & Active Learning Approaches	16	6
Total		52	55

The distribution indicates that a substantial proportion of studies, both in India and abroad, are concentrated in areas such as ICT integration, blended learning, and general active learning approaches. While these studies contribute to improving engagement and learning outcomes, comparatively fewer studies explicitly focus on micro-learning and connectivist approaches, particularly in combination. Further, international research demonstrates a relatively stronger orientation towards intervention-based designs, whereas Indian studies are still developing in terms of structured implementation and empirical validation of such approaches.

Across both contexts, a common limitation is the lack of integrated instructional frameworks that simultaneously address micro-learning design, connectivist interaction, and the development of critical thinking within discipline-specific contexts such as physics. In addition, there is limited use of domain-specific assessment approaches to objectively evaluate critical thinking skills in such settings.

Research Gap

A careful examination of the reviewed literature reveals several specific gaps that the present study seeks to address:

1. **Subject-Specific Gap:** While micro-learning has been explored in areas such as language learning, management education, and school-level instruction, its application in undergraduate physics—particularly for conceptually demanding

topics—remains limited. There is a lack of studies focusing on the development of critical thinking within such abstract scientific domains using structured micro-learning approaches.

2. **Pedagogical Integration Gap:** Existing studies tend to examine micro-learning, connectivist learning environments, and critical thinking as separate constructs. There is limited research that integrates these elements into a unified instructional framework aimed at systematically enhancing higher-order cognitive skills.
3. **Methodological Gap:** A substantial proportion of studies, particularly in the Indian context, are descriptive or survey-based. There is a need for intervention-based research designs that empirically evaluate instructional effectiveness. Additionally, the use of domain-specific assessment tools to objectively measure critical thinking in physics remains limited.
4. **Contextual Gap:** Many technology-supported learning models reported in the literature are developed in well-resourced environments. There is comparatively limited research focusing on the design and implementation of accessible, low-cost digital learning approaches suitable for resource-constrained undergraduate settings.

The present study addresses these gaps by developing, implementing, and evaluating a connectivist micro-learning programme for undergraduate physics students, with a focus on enhancing and assessing critical thinking within a specific disciplinary context.

7 Objectives of the Study

The present study has been undertaken with the following specific objectives:

1. To study connectivist principles relevant to the design of micro-learning activities aimed at enhancing critical thinking skills in undergraduate physics students.
2. To identify and validate connectivist principles for the development of the micro-learning programme through expert review.
3. To conduct a needs analysis survey of undergraduate physics teachers to identify the topics in the First-Year undergraduate physics syllabus (NEP 2020) that are perceived as challenging to teach, difficult for students to learn, and requiring instructional support through micro-learning activities.
4. To design and develop a connectivist micro-learning programme based on the validated connectivist principles and the identified difficult topic in undergraduate physics.

5. To implement and evaluate the effectiveness of the developed micro-learning programme for enhancing the critical thinking skills of undergraduate physics students.
6. To give evidence-based implications and suggestions for curriculum planners, institutions, teachers, and future researchers based on the findings of the study.

8 Assumptions of the Study

The present study is based on the following foundational assumptions:

1. Critical thinking is a cognitive skill that can be meaningfully fostered, developed, and assessed within higher education contexts through appropriately designed instructional experiences (Ennis, 2011; Facione, 1990).
2. Micro-learning is a pedagogical approach that is theoretically suited to addressing cognitive load, sustaining learner engagement, and maintaining attention when dealing with complex conceptual content (Giurgiu, 2017; Nikou & Economides, 2018).
3. Connectivist principles provide a relevant and appropriate theoretical framework for designing and facilitating learning in digitally mediated, networked environments commonly used in contemporary higher education (Downes, 2007; Siemens, 2005).
4. Students and teachers participating in the study will engage with the instructional activities and assessment procedures in good faith, in accordance with typical academic practices and the ethical norms of educational research (Cohen et al., 2018).

9 Hypotheses of the Study

In order to empirically fulfill Objective 5 of the present study, which seeks to evaluate the effectiveness of the developed micro-learning programme, the following null (H_0) and alternative (H_1) hypotheses were formulated:

- H_0 There is no significant difference in the overall critical thinking skills of undergraduate physics students between the experimental and control groups following the implementation of the connectivist micro-learning programme.
- H_1 There is a significant difference in the overall critical thinking skills of undergraduate physics students between the experimental and control groups following the implementation of the connectivist micro-learning programme.

10 Significance of the Study

The present study is significant in the context of contemporary higher education reforms in India, particularly in alignment with the National Education Policy (NEP) 2020 (Ministry of Education, 2020). NEP 2020 emphasises the development of higher-order cognitive skills such as critical thinking, problem-solving, and learner autonomy. However, in undergraduate physics education, these goals remain difficult to achieve, as traditional teaching methods often prioritise formula-based problem solving over conceptual understanding and reasoning.

The significance of the study is further grounded in local educational realities. Interactions with physics teachers from affiliated colleges in Kolhapur city and from relevant departments within Shivaji University indicate challenges such as low student engagement, irregular attendance, and difficulty in understanding abstract topics. A needs analysis survey conducted among these teachers identified *Electrostatics* as one of the most difficult topics for students. By focusing on this topic, the study addresses a classroom-relevant instructional problem.

From a practical perspective, the study develops a connectivist micro-learning programme that can be used by teachers in undergraduate classrooms. The programme organises content into short, interaction-based learning activities delivered through a digital platform (Slack). This provides a flexible and accessible approach that is suitable for implementation in online or blended learning environments without requiring complex technological infrastructure.

From an academic perspective, the study contributes to the existing literature by integrating micro-learning, connectivist principles, and critical thinking within a single instructional approach. While these areas have been studied separately (Giurgiu, 2017; Siemens, 2005; Tiruneh et al., 2017), their combined application in undergraduate physics education, particularly in the Indian context, remains limited.

Overall, the study contributes to physics pedagogy, aligns with national educational priorities, and offers a structured approach for enhancing critical thinking skills among undergraduate students.

11 Scope of the Study

The scope of the present study lies within undergraduate physics education at the First Year B.Sc. level in colleges affiliated with Shivaji University, Kolhapur, as well as relevant departments within the Shivaji University campus where physics is taught.

The study focuses on the design, development, and implementation of a connectivist micro-learning programme for teaching a conceptually challenging topic in physics, namely *Electrostatics*. It explores the use of structured, interaction-driven micro-learning activities as an alternative to traditional instructional approaches.

The scope of the study includes the enhancement and assessment of students' critical thinking skills within the context of undergraduate physics learning.

The study also includes the use of digital platforms to support learning through discussion, interaction, and collaborative engagement in an online environment.

The findings and instructional approach of the study are expected to be applicable to similar undergraduate teaching contexts where comparable academic conditions and classroom-based instructional settings exist.

12 Delimitations of the Study

The present study is delimited as follows:

1. The study is delimited to Kolhapur City, which lies within Karvir Taluka of Kolhapur District.
2. The study is delimited to colleges affiliated to Shivaji University, Kolhapur, within Kolhapur City, and to the departments of Shivaji University, Kolhapur, where undergraduate physics is offered.
3. The study is delimited to physics teachers teaching First Year B.Sc. Physics courses in affiliated colleges and in the departments of Shivaji University, Kolhapur, for the purpose of the needs analysis survey.
4. The study is delimited to undergraduate students of First Year B.Sc. from one selected department of Shivaji University, Kolhapur, for the pilot study, out of the defined study population.
5. The study is delimited to the implementation of the connectivist micro-learning programme within one selected department of Shivaji University, Kolhapur, for the main experimental study, out of the defined study population.
6. The study is delimited to the subject of Physics, focusing on a single topic, namely Electrostatics, as identified through the needs analysis survey.
7. The study is delimited to the academic year 2025–2026.
8. The study is delimited to the use of a quantitative research method, with no qualitative data collection or analysis undertaken.
9. The study is delimited to the assessment of critical thinking skills using a structured assessment framework.
10. The study is delimited to the development and evaluation of a single researcher-developed connectivist micro-learning programme and does not include comparison with alternative instructional approaches or existing learning platforms.

11. The study is delimited to the use of the Slack platform as the medium for delivering the micro-learning programme.

13 Limitations of the Study

The present study was subject to the following limitations:

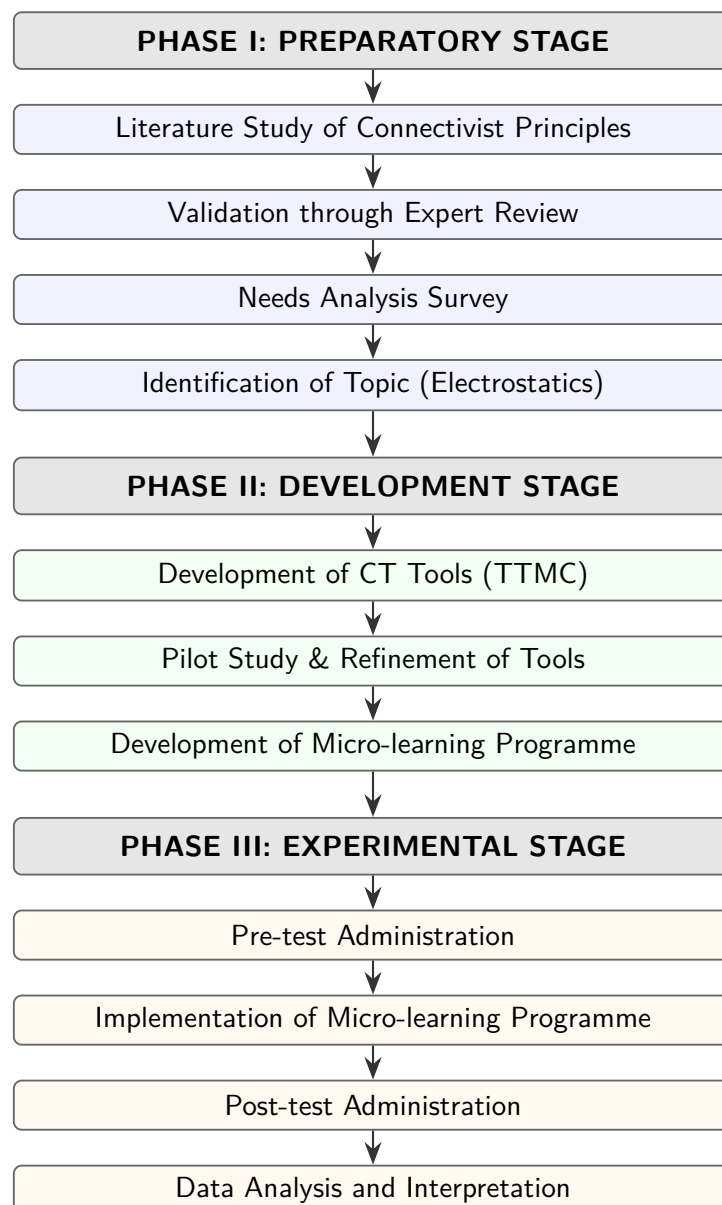
1. The study was conducted using intact groups, as random assignment of students was not feasible due to practical constraints related to classroom organisation and variability in student attendance.
2. The main experimental study was conducted within a single selected institutional setting affiliated to Shivaji University, Kolhapur, which may limit the generalisability of the findings.
3. Participation across different phases of the study varied. Although enrolment numbers were higher, student attendance during the pre-test and post-test was not consistent. Consequently, only complete and valid responses were retained, resulting in a reduced final sample size.
4. The implementation of the micro-learning programme was affected by constraints related to student access and participation, including issues of internet connectivity and varying levels of familiarity with the digital platform.
5. The duration of the intervention was reduced from the originally planned schedule due to administrative constraints and interruptions during implementation.
6. The assessment tools used in the study were researcher-developed. Although they were validated and refined through pilot testing, certain measurement limitations may persist.
7. The assessment of critical thinking was based on two-tier multiple-choice instruments. While this format captures reasoning, it may not fully reflect the depth of students' thinking compared to open-ended responses.
8. Advanced psychometric analyses, such as confirmatory factor analysis, were not undertaken due to limitations in sample size.
9. External factors such as prior knowledge, student motivation, and learning environment could not be fully controlled.

14 Procedure of the Research

The present study was carried out in a systematic sequence of phases, beginning with the identification of relevant connectivist principles and concluding with the analysis and interpretation of data. The overall procedure followed in the study is presented in Figure 1.

Figure 1

Flowchart of the Phased Research Procedure



15 Sampling Design and Sample

15.1 Population Framework of the Study

The present study involved three distinct populations corresponding to different phases of the research process. These populations were identified based on the objectives of the study and the stages of data collection.

1. **Institutional Population:** This included all colleges affiliated to Shivaji University, Kolhapur, and relevant departments within the university that offered First-Year B.Sc. Physics under the NEP 2020-aligned curriculum within Kolhapur City.
2. **Teacher Population:** This consisted of physics teachers teaching First-Year B.Sc. Physics in the identified institutions. This population was relevant for the needs analysis survey.
3. **Student Population:** This included First-Year B.Sc. students studying physics in the identified institutions. This population was relevant for the pilot study and the main experimental study.

These three populations formed the basis for subsequent sampling procedures carried out at different stages of the study, as summarized in Table 3.

Table 3
Population Framework of the Study

Sr. No.	Population Type	Purpose in Study
1	Institutions (Colleges and University Departments)	Define study setting
2	Physics Teachers (F.Y. B.Sc.)	Needs Analysis Survey
3	Undergraduate Physics Students (F.Y. B.Sc.)	Pilot and Experimental Study

15.2 Population of Institutions

The institutional population of the present study consisted of all institutions within Kolhapur City (Karveer Taluka, Kolhapur District, Maharashtra) that met the following criteria:

- Affiliation to Shivaji University, Kolhapur.
- Implementation of the NEP 2020–aligned undergraduate physics curriculum for First-Year B.Sc. during the academic years 2024–2025 and 2025–2026.

Based on these criteria, a total of seven institutions were identified, as detailed in Table 4. This included five affiliated colleges and two departments within Shivaji University, Kolhapur, where First-Year B.Sc. Physics is offered. No institution within the defined geographical and academic scope that met the above criteria was omitted from the study.

Table 4
Institutional Population of the Study

Sr. No.	Institution	Type
1	Rajaram College	Affiliated College
2	Shahu College	Affiliated College
3	The New College	Affiliated College
4	Shahaji College	Affiliated College
5	Gokhale College	Affiliated College
6	Department of Education, SUK	University Department
7	Dept. of Nanoscience and Technology, SUK	University Department

Thus, the institutional population comprised a complete set of all eligible colleges and departments within the defined scope of the study.

15.3 Teacher Population and Sample (Needs Analysis)

The teacher population and sample for the needs analysis survey were defined as follows:

1. **Teacher Population:** The population consisted of all physics teachers teaching First-Year B.Sc. Physics in the identified institutions under the NEP 2020–aligned curriculum.
2. **Inclusion Criteria:** Only those teachers who were actively teaching First-Year B.Sc. Physics at the time of data collection were considered for inclusion.

3. **Exclusion Criteria:** Teachers who were teaching physics at higher undergraduate levels but were not involved in First-Year B.Sc. instruction were excluded, as the focus of the study was specifically on the First-Year syllabus.
4. **Final Sample:** Based on the above criteria, a total of 19 teachers from the identified institutions participated in the needs analysis survey, distributed as shown in Table 5.
5. **Nature of Sampling:** Since all eligible teachers within the defined population were included, the needs analysis survey can be considered a census of the accessible population.

Table 5*Distribution of Teachers Participating in Needs Analysis*

Sr. No.	Institution	No. of Teachers
1	Rajaram College	4
2	Shahu College	1
3	The New College	5
4	Shahaji College	1
5	Gokhale College	4
6	Department of Education, SUK	0
7	Dept. of Nanoscience and Technology, SUK	4
Total		19

At the time of data collection, the Department of Education did not have a physics teacher assigned to First-Year B.Sc. Physics and was therefore not represented in the needs analysis survey.

15.4 Student Population

The student population of the present study consisted of all First-Year B.Sc. students enrolled in undergraduate physics courses in the identified institutions under the NEP 2020-aligned curriculum. However, it was observed that regular classroom attendance varied considerably across institutions, and not all enrolled students were consistently

present during instructional sessions. Therefore, the selection of students for the pilot and experimental phases was based on actual availability and participation during the data collection period.

15.5 Sample for Pilot and Experimental Study

Student samples for the present study were selected in two stages, with institutional utilization mapped in Table 6:

1. **Pilot Study Sample:** The pilot study was conducted on First-Year B.Sc. students from the Department of Education, Shivaji University, Kolhapur. A total of 26 students participated in the pilot study.
2. **Main Experimental Study Sample:** The main study was conducted using intact groups and comprised:
 - **Experimental Group:** Students from the Department of Nanoscience and Technology, Shivaji University, Kolhapur.
 - **Control Group:** Students from selected affiliated colleges, namely Rajaram College, Gokhale College, and The New College.

Table 6

Utilisation of Institutions for Student Sample

Sr. No.	Institution	Pilot Study	Main Study	
			Exp. Group	Control Group
1	Rajaram College			✓
2	Shahu College			
3	The New College			✓
4	Shahaji College			
5	Gokhale College			✓
6	Department of Education, SUK	✓		
7	Dept. of Nanoscience and Tech, SUK		✓	

During the pre-test phase, 56 students from the experimental group and 43 students from the control group participated. After data screening—which included retaining

only those students who participated in both the pre-test and post-test and provided valid responses—the final sample consisted of 27 students in the experimental group and 34 students in the control group. The numerical breakdown across these stages is provided in Table 7.

Table 7

Student Participation Across Different Stages of the Study

Study Stage	Group	Number of Students
Pilot Study	Department of Education	26
Main Study (Pre-test)	Experimental Group (Nanoscience Dept.)	56
	Control Group (3 Colleges)	43
Main Study (Final)	Experimental Group	27
	Control Group	34

15.6 Sampling Technique

The sampling process for the present study was carried out in multiple stages using appropriate techniques at each level:

1. **Institutional Level:** Purposive sampling was used to identify institutions based on predefined criteria, namely affiliation to Shivaji University, Kolhapur, and implementation of the NEP 2020–aligned undergraduate physics curriculum.
2. **Teacher Level (Needs Analysis):** A census approach was adopted, wherein all accessible physics teachers teaching First-Year B.Sc. Physics were included in the survey.
3. **Student Level:**
 - **Pilot Study:** Students were selected using purposive sampling based on accessibility and feasibility.
 - **Main Experimental Study:** Purposive sampling was used to select institutions and student groups based on the feasibility of participation. Intact classroom groups were utilized, as random assignment of individual students was not feasible.

The overall sampling process thus involved a combination of purposive sampling, the census method, and intact group selection within a **pre-test–post-test non-equivalent control group design**(quasi-experimental framework).

16 Tools for Data Collection

The present study utilised researcher-developed tools across different stages of the research process, namely validation, needs analysis, and assessment of critical thinking skills.

1. **Connectivist Principles Checklist (Validation Tool):** A checklist was developed by the researcher to identify relevant connectivist principles for the design of the micro-learning programme. The checklist was subjected to expert review to ensure the relevance and appropriateness of the selected principles.
2. **Needs Analysis Questionnaire:** A structured questionnaire was developed and administered to physics teachers teaching First-Year B.Sc. Physics. The tool was used to identify topics that are challenging to teach, difficult for students to learn, and suitable for instructional support through micro-learning activities.
3. **Critical Thinking Assessment Tools:**
 - **Paper 1 (GCTA): General Critical Thinking Assessment** A researcher-developed two-tier multiple-choice assessment designed to measure the overall critical thinking skills of undergraduate students.
 - **Paper 2 (PGCTA): Physics Domain-Specific Critical Thinking Assessment** A researcher-developed two-tier multiple-choice assessment designed to measure critical thinking skills within the context of physics, specifically focusing on the topic of Electrostatics.

All the tools were subjected to expert validation to ensure content relevance and appropriateness. The critical thinking assessment tools were further refined through pilot testing to ensure clarity and suitability before their use in the main experimental study.

17 Development of the Micro-learning Programme

The connectivist micro-learning programme developed in the present study was designed as an interaction-driven, discussion-based learning environment aimed at enhancing the critical thinking skills of undergraduate physics students.

1. **Basis of Development:** The programme was developed systematically based on validated connectivist principles, findings from the needs analysis survey, and the identification of a difficult topic in the First-Year undergraduate physics syllabus.
2. **Design of the Programme:** The programme was structured to promote active engagement, reasoning, and interaction. It emphasised peer discussion, exposure to multiple perspectives, and continuous engagement with concepts rather than passive content reception.
3. **Platform and Mode of Delivery:** The programme was implemented through the Slack digital platform, which enabled threaded discussions, asynchronous participation, and resource sharing among students.
4. **Structure of Learning Activities:** Each learning session was initiated through a trigger in the form of a question or problem. Students responded with answers and justifications, followed by peer interaction through discussion and comparison of reasoning.

The overall structure followed a recurring engagement cycle consisting of:

- Initiation through a trigger
- Individual response
- Peer interaction
- Guided facilitation
- Reflective observation
- Consolidation of ideas

5. **Micro-learning Implementation:** Micro-learning was operationalised through short, focused sessions, each centred on a specific concept within the topic of Electrostatics. The programme was conducted over approximately one week, including an initial orientation session followed by concept-based learning sessions, as detailed in Table 8.

Table 8
Session-wise Structure of the Micro-learning Programme

Date	Session Type	Topic / Focus	Nature of Activity
02-03-2026	Orientation	Introduction to Slack	Platform familiarisation and participation guidelines
04-03-2026	Session 1	Coulomb's Law	Trigger-based conceptual question, individual response, peer discussion
05-03-2026	Session 2	Electric Field	Problem-based prompt, justification of answers, peer interaction
06-03-2026	Session 3	Electric Field Lines	Conceptual reasoning task, comparison of responses, discussion
07-03-2026	Session 4	Gauss's Law	Application-based problem, critical reasoning, guided facilitation and consolidation

Each session was conducted within a fixed engagement window from 6:30 p.m. to 7:15 p.m., as mutually agreed upon with students to facilitate consistent participation.

6. **Role of the Researcher and Teacher:** The researcher functioned as the primary facilitator, responsible for initiating discussions, posing probing questions, and guiding student interaction without direct content delivery. The classroom teacher was included in the platform for supervision.
7. **Connectivist Features of the Programme:** The programme emphasised learning through interaction, encouraging students to engage with peer responses, utilise multiple information sources, and participate in structured discussion threads.
8. **Validation of the Programme:** The developed micro-learning programme was subjected to content validation by a panel of experts. The Content Validity Index (CVI) indicated a high level of agreement among experts, and necessary modifications were incorporated prior to implementation. The overall S-CVI/Ave value of the programme was 0.90, indicating high content validity.

18 Analysis, Interpretation of Data and Results

The data collected in the present study were analysed in accordance with the stated objectives. The analysis and interpretation were carried out in the following stages:

1. **Analysis of Connectivist Principles:** The connectivist principles identified through literature review were subjected to expert validation. A total of 15 connectivist principles were evaluated by 7 experts.

The analysis indicated that all 15 principles were considered relevant and appropriate based on majority agreement. These validated principles formed the basis for the design of the micro-learning programme.

2. **Analysis of Needs Analysis Teacher Survey:** The responses obtained from undergraduate physics teachers were analysed using frequency, percentage, and ranking methods (Borda count) to identify topics suitable for micro-learning intervention.

The findings revealed that *Electrostatics* received the highest ranking across multiple criteria, including topics that are challenging to teach, require critical thinking, and are suitable for micro-learning.

Further analysis indicated that the primary difficulty reported by teachers was the application of concepts to real-world situations. Based on these findings, *Electrostatics* was selected as the focus of the micro-learning programme.

3. **Pilot Study Analysis:** The pilot study was conducted to examine the clarity and functioning of the assessment tools.

The results indicated low baseline performance among students, with mean scores of 3.58 (17.90%) for Paper 1 (GCTA) and 3.19 (15.95%) for Paper 2 (PGCTA).

The alternate forms of the tests were found to be comparable in difficulty, showing no significant difference between forms. Reliability estimates were moderate for Paper 1 ($\alpha = 0.665$) and low for Paper 2 ($\alpha = 0.201$), which was interpreted in light of the small sample size and the stringent two-tier scoring format.

Based on item analysis and observed response patterns, necessary modifications were made to improve clarity and functioning of the tools.

4. **Main Study Analysis:** The main experimental study involved pre-test and post-test assessments for experimental and control groups.

- Descriptive analysis indicated higher post-test performance in the experimental group compared to the control group for both assessments.
- Independent samples comparisons and non-parametric alternatives were used where necessary to examine differences between groups.

- Analysis of Covariance (ANCOVA) was conducted to control for pre-test differences and obtain a more precise estimate of the intervention effect.
- ANCOVA results indicated a statistically significant difference in favour of the experimental group for both Paper 1 (GCTA) ($p = 0.022$) and Paper 2 (PGCTA) ($p = 0.002$).

5. Testing of Hypothesis:

H_0 : There is no significant difference in the overall critical thinking skills of undergraduate physics students between the experimental and control groups.

H_1 : There is a significant difference in the overall critical thinking skills of undergraduate physics students between the experimental and control groups.

Decision: Analysis of Covariance (ANCOVA) revealed a statistically significant difference in favour of the experimental group for both Paper 1 (GCTA) ($F(1, 57) = 5.53, p = 0.022$) and Paper 2 (PGCTA) ($F(1, 57) = 10.68, p = 0.002$). Therefore, the null hypothesis (H_0) is rejected and the alternative hypothesis (H_1) is accepted.

Conclusion: The connectivist micro-learning programme was found to be effective in enhancing the critical thinking skills of undergraduate physics students.

Objective-wise Summary of Findings

Table 9

Objective-wise Summary of Findings

Objective	Key Findings & Evidence
Objective 1	Relevant connectivist principles for designing micro-learning activities were identified through an extensive review of literature (Downes, 2007; Siemens, 2005). A total of 15 principles were shortlisted for further validation.
Objective 2	All 15 connectivist principles were validated through expert review ($N = 7$). Based on majority agreement, all principles were considered relevant and appropriate for inclusion in the programme design.

Continued on next page

Table 9 continued from previous page

Objective	Key Findings & Evidence
Objective 3	Identified Difficult Topic: Electrostatics received the highest ranking across criteria including difficulty in teaching, requirement of critical thinking, and suitability for micro-learning. Pedagogical Difficulty: The most reported challenge was application of concepts to real-world scenarios.
Objective 4	A connectivist micro-learning programme was designed and developed based on validated principles and identified topic. The programme included structured discussion-based activities, peer interaction, and resource sharing through a digital platform (Slack), and was validated using Content Validity Index (CVI).
Objective 5	Paper 1 (GCTA): ANCOVA indicated a significant difference in favour of the experimental group ($F(1, 57) = 5.53, p = 0.022, \eta_p^2 = 0.088$). Paper 2 (PGCTA): ANCOVA indicated a significant difference in favour of the experimental group ($F(1, 57) = 10.68, p = 0.002, \eta_p^2 = 0.158$). Interpretation: The micro-learning programme was effective in enhancing both general and physics-specific critical thinking skills.
Objective 6	Based on the findings of the study, evidence-based implications and recommendations were formulated for teachers, curriculum planners, and future researchers.

19 Recommendations / Implications of the Findings

Based on the findings of the present study, the following recommendations and implications are proposed:

1. Implications for Teachers

- The findings indicate that micro-learning strategies may be incorporated into classroom practice, particularly for conceptually difficult topics such as Electrostatics. The observed improvement in the experimental group suggests that short, structured learning activities can support the development of critical thinking.
- The effectiveness of the connectivist micro-learning programme highlights the value of discussion-based and interaction-driven instructional approaches. Teachers may therefore include structured opportunities for peer interaction and collaborative reasoning alongside conventional teaching methods.
- The variation in improvement across different critical thinking constructs suggests that critical thinking develops unevenly. Instructional strategies may therefore include targeted activities focusing on specific components such as evaluating evidence, identifying reasoning flaws, and analysing relationships between variables.
- The use of a digital platform in the present study indicates that technology may be integrated into teaching to extend learning beyond the classroom, particularly in contexts with limited instructional time.

2. Implications for Curriculum Planners and Institutions

- The findings suggest that structured micro-learning approaches may support the development of critical thinking skills in undergraduate physics. Curriculum planners may therefore consider integrating such approaches within existing courses, particularly for conceptually challenging topics.
- The implementation of digital learning interventions requires institutional support. The study highlights the importance of ensuring access to reliable internet connectivity and appropriate technological infrastructure for both teachers and students.
- There is a need for professional development programmes to familiarise teachers with the design and implementation of micro-learning and connectivist instructional approaches.

20 Suggestions for Further Research

Based on the findings and limitations of the present study, the following suggestions are proposed for future research:

1. Future studies may be conducted with larger and more diverse samples to enhance generalisability and improve statistical stability.
2. Longitudinal studies may be undertaken to examine the sustained impact of micro-learning interventions on the development of critical thinking over time.
3. Further research may focus on individual constructs of critical thinking to design targeted interventions, particularly for constructs that did not show significant improvement.
4. Studies may be conducted to examine the transfer of critical thinking skills across domains under varied instructional conditions.
5. Future research may incorporate mixed-method approaches by combining quantitative assessments with qualitative measures such as reflective responses to gain deeper insights into students' thinking processes.
6. Similar studies may be conducted in different subjects and educational contexts to examine the applicability of connectivist micro-learning approaches beyond physics education.
7. Further research may investigate the influence of contextual factors such as access to technology, student participation patterns, and learning environments on the effectiveness of digital instructional interventions.
8. Future studies may explore the implementation of micro-learning interventions using alternative digital platforms, such as messaging-based applications (e.g., WhatsApp), to examine their feasibility and effectiveness in diverse educational contexts.